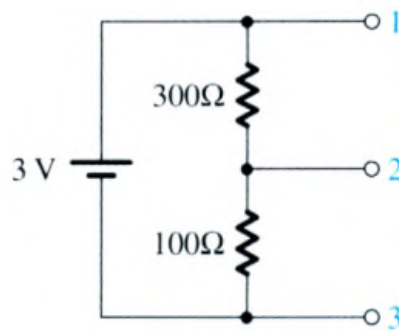


University of Wisconsin-Madison
Department of Electrical and Computer Engineering
ECE 340 - Electronic Circuits I, Fall Semester 2026
Homework #1

Submission deadline: Tuesday, September 15, 2026, at 11:59 pm.

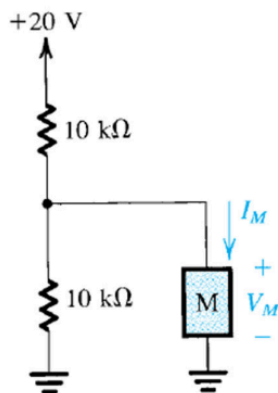
Problem 1

1.15 For the circuit in Fig. P1.15, find the Thévenin equivalent circuit between terminals (a) 1 and 2, (b) 2 and 3, and (c) 1 and 3.



Problem 2

The two-terminal device M in the circuit below has a current $I_M \approx 1\text{mA}$ independent of the voltage V_M across it. Use Norton's theorem to simplify the circuit to the left of M and draw the equivalent circuit. Calculate voltage V_M using the equivalent circuit.



(Ans. 5V)

Problem 3

1.46 An amplifier operating from $\pm 3\text{-V}$ supplies provides a 2.2-V peak sine wave across a $100\text{-}\Omega$ load when provided with a 0.2-V peak input from which 1.0 mA peak is drawn. Find the voltage gain, current gain, and power gain expressed as ratios and in decibels. If the amplifier efficiency is 10% , find the supply power, supply current, and amplifier dissipation.

Problem 4

1.53 You are given two amplifiers, A and B, to connect in cascade between a 10-mV , $100\text{-k}\Omega$ source and a $100\text{-}\Omega$ load. The amplifiers have voltage gain, input resistance, and output resistance as follows: for A, 100 V/V , $100\text{ k}\Omega$, $10\text{ k}\Omega$, respectively; for B, 10 V/V , $10\text{ k}\Omega$, $1\text{ k}\Omega$, respectively. Your problem is to decide how the amplifiers should be connected. To proceed, evaluate the two possible connections between source S and load L, namely, SABL and SBAL. Find the voltage gain for each both as a ratio and in decibels. Which amplifier arrangement is best?

Problem 5

- 1.17** (a) Model the three-stage amplifier of Example 1.3 (without the source and load), using the voltage amplifier model. What are the values of R_i , A_{vo} , and R_o ?
 (b) If R_L varies in the range $10\ \Omega$ to $1000\ \Omega$, find the corresponding range of the overall voltage gain, v_o/v_s .

Ans. $1\ \text{M}\Omega$, $900\ \text{V/V}$, $10\ \Omega$; $409\ \text{V/V}$ to $810\ \text{V/V}$

Example 1.3

Figure 1.17 depicts an amplifier composed of a cascade of three stages. The amplifier is fed by a signal source with a source resistance of $100\ \text{k}\Omega$ and delivers its output into a load resistance of $100\ \Omega$. The first stage has a relatively high input resistance and a modest gain factor of 10. The second stage has a higher gain factor but lower input resistance. Finally, the last, or output, stage has unity gain but a low output resistance. We wish to evaluate the overall voltage gain, that is, v_L/v_s , the current gain, and the power gain.

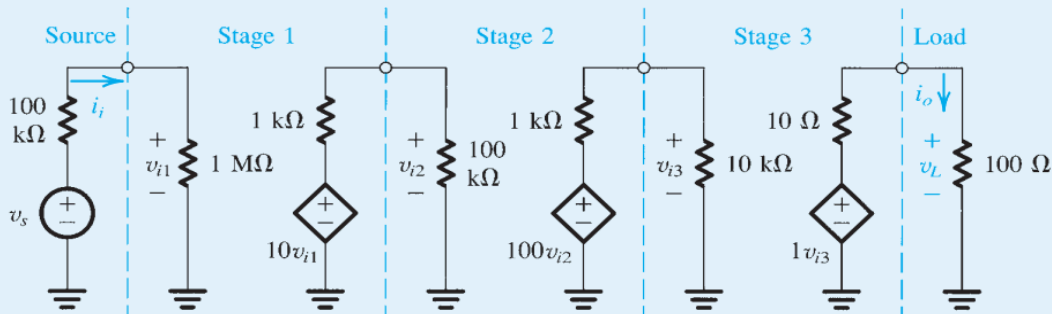


Figure 1.17 Three-stage amplifier for Example 1.3.